

Arbaeen Pilgrim Assistant Chatbot

An AI-powered conversational assistant to support pilgrims during the Arbaeen walk from Najaf to Karbala, Iraq

Student	Student ID	Methodology	Sprints	Total Story Points
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Scan to try the live chatbot



PROBLEM STATEMENT

Arbaeen is one of the largest annual human gatherings on earth, drawing an estimated 20 to 25 million pilgrims to Karbala, Iraq each year. Pilgrims walking the approximately 80-kilometer route from Najaf to Karbala frequently face critical challenges: locating Mawkib food and rest stations, handling heat exhaustion and dehydration, treating foot injuries, and finding emergency contacts. Access to timely, accurate, and accessible guidance during the walk remains a genuine and unresolved problem for millions of participants every year.

TECHNOLOGY STACK

FRONTEND

React.js (Vite)

AI MODEL

Llama 3.1 8B

API

Groq (Free Tier)

HOSTING

Vercel

VERSION CONTROL

Git and GitHub

AI DEV TOOLS

Claude, ChatGPT

CORE OBJECTIVES



Mawkib Guidance

Help pilgrims locate food, water, and rest stations along the Najaf to Karbala route through natural language queries



First Aid Support

Provide instant guidance for common health issues including heat exhaustion, dehydration, and foot blisters



Route Assistance

Answer questions about walking paths, distances, checkpoints, and emergency contact numbers during the walk

USER STORIES

US07 As a pilgrim, I want a simple chat interface on my phone so that I can type my question and get an instant answer.

5 pts

High

US01 As a pilgrim, I want to ask about Mawkib stations along the route so that I can find food and rest during my walk.

3 pts

High

US04 As a pilgrim, I want to ask what to do if I feel dizzy or overheated so that I can handle heat exhaustion quickly.

3 pts

High

US02 As a pilgrim, I want to ask about the distance remaining to Karbala so that I can plan my walking pace.

2 pts

High

US08 As a pilgrim, I want to see suggested quick questions when I open the app so that I know what I can ask without typing.

3 pts

Medium

US11 As a pilgrim, I want to ask about emergency contact numbers in Iraq so that I can call for help if needed.

1 pt

Low

MILESTONES AND ROADMAP

- Sprint 0: Setup**
Environment setup, Groq API integration, React project initialized
- Sprint 1: Core Features**
Chatbot UI built, system prompt designed, Mawkib and first aid guidance active
- Sprint 2: Enhancements**
Quick questions added, conversation memory, mobile responsiveness complete
- Sprint 3: MVP Complete**
Emergency contacts, deployed on Vercel, GitHub repository published

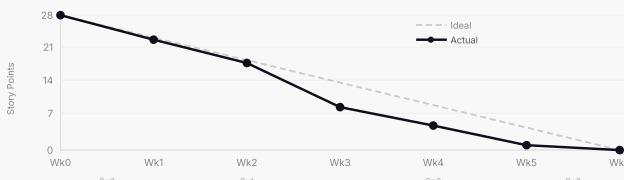
SPRINT BOARD

Sprint	To Do	In Progress	Done
Sprint 0 Setup			US07: Chat interface and Groq API (5 pts)
Sprint 1 Core			US01, US02, US03, US04, US05, US06 (14 pts)
Sprint 2 Enhance			US08, US09, US10 (8 pts)
Sprint 3 Polish			US11: Emergency contacts (1 pt)

VELOCITY CHART

Sprint 0	5
Sprint 1	14
Sprint 2	8
Sprint 3	
Total	28 pts

BURNDOWN CHART



CHALLENGES AND NEEDS

- Designing a system prompt that keeps AI responses focused strictly on Arbaeen-related topics while remaining helpful and accurate.
- Managing API rate limits on the Groq free tier to ensure reliable performance during demonstration and actual use.
- Building a mobile-first interface that remains usable under stress conditions with minimal interaction steps required from the pilgrim.
- Balancing a realistic project scope within a 6-week academic timeline while delivering a functional and demonstrable product.